

Exercise 1: SELECT and Filtering

SQL Fundamentals

Table 1: employees

01 Selecting & Sorting

Start with the basics: pulling data out of the table, removing duplicates, sorting it, and limiting how many rows you get back. The **employees** table below is the data source for every question in this exercise.

TABLE employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Question 1

01 Retrieve all columns from the employees table.

EXPECTED COLUMNS: **id** **first_name** **last_name** **department** **salary**
hire_date **city**

SQL code:

▶ Run all (1000) ✓ Just now (14s) workspace default Edit

```
1 --Question 1: Retrieve all columns from the employees table, EXPECTED COLUMNS: id
  first_name, last_name, department, salary, hire_date city
2 SELECT *
3 FROM brightlearn.master_data.employees;
4
```

Add parameter

Table	id	first_name	last_name	department	salary	hire_date
1	1	Alice	Green	IT	70000	2020-01-10
2	2	Brian	Lee	HR	45000	2019-03-22
3	3	Cathy	Zulu	Finance	65000	2018-07-18
4	4	David	Mokoena	Marketing	50000	2021-11-05
5	5	Eva	Naidoo	IT	72000	2017-09-30

Question 2

02 Find all unique departments.

EXPECTED COLUMNS: **department**

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace default ▾

```
5  --Question 2
6  SELECT DISTINCT department
7  FROM brightlearn.master_data.employees;
8
```

Add parameter

Table ▾ +

	department
1	IT
2	HR
3	Finance
4	Marketing

Question 3

03 Retrieve first and last names ordered by salary descending.

EXPECTED COLUMNS: **first_name** **last_name**

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace default ▾

```
8
9  --Question 3
10 SELECT first_name,
11         last_name
12 FROM brightlearn.master_data.employees
13 ORDER BY salary DESC;
14
```

Add parameter

Table ▾ +

	first_name	last_name
1	Eva	Naidoo
2	Alice	Green
3	Cathy	Zulu
4	David	Mokoena
5	Brian	Lee

Question 4

04

Retrieve the top 3 highest-paid employees.

EXPECTED COLUMNS:

id

first_name

last_name

salary

SQL code:

▶ Run all (1000) ▼

✓ Just now (1s) workspace: default ▼

```
--Question 4
SELECT id, first_name,
       last_name,
       salary
FROM brightlearn.master_data.employees
ORDER BY SALARY DESC
LIMIT 3;
```

Add parameter

Table ▼



	¹ ₃ id	^A _C first_name	^A _C last_name	¹ ₃ salary
1	5	Eva	Naidoo	72000
2	1	Alice	Green	70000
3	3	Cathy	Zulu	65000

Question 5

05

Find employees in the IT department.

EXPECTED COLUMNS:

id

first_name

last_name

department

SQL code:

▶ Run all (1000) ▼

✓ Just now (1s) workspace: default ▼

```
--Question 5
SELECT id,
       first_name,
       last_name,
       department
FROM brightlearn.master_data.employees
WHERE department = 'IT';
```

Add parameter

Table ▼



	¹ ₃ id	^A _C first_name	^A _C last_name	^A _C department
1	1	Alice	Green	IT
2	5	Eva	Naidoo	IT

Question 6:

06 Find employees in Finance with salary > 60000.

EXPECTED COLUMNS: id first_name last_name department salary

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace. default ▾

```
30
31 --Question 6
32 SELECT id,
33        first_name,
34        last_name,
35        department,
36        salary
37 FROM brightlearn.master_data.employees
38 WHERE salary > 60000 AND department = 'Finance';
39
```

Add parameter

Table ▾ +		Q Y I O E ✓			
	¹ ₃ id	^A _C first_name	^A _C last_name	^A _C department	¹ ₃ salary
1	3	Cathy	Zulu	Finance	65000

Question 7

07 Find employees in HR or Marketing.

EXPECTED COLUMNS: id first_name last_name department

SQL code:

▶ Run all (1000) ✓ Just now (<1s) workspace. default ▾

```
39
40 --Question 7
41 SELECT id,
42        first_name,
43        last_name,
44        department
45 FROM brightlearn.master_data.employees
46 WHERE department = 'HR'
47 OR
48        department = 'Marketing';
```

Add parameter

Table ▾ +		Q		
	¹ ₃ id	^A _C first_name	^A _C last_name	^A _C department
1	2	Brian	Lee	HR

Question 8

08 Find employees not in IT.

EXPECTED COLUMNS: id first_name last_name department

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace. default ▼

```
49
50 --Question 8
51 SELECT id,
52        first_name,
53        last_name,
54        department
55 FROM brightlearn.master_data.employees
56 WHERE department <> 'IT';
57
```

Add parameter

Table ▼ +					Q 🔍	
	1 ² 3 id	A ^B C first_name	A ^B C last_name	A ^B C department		
1	2	Brian	Lee	HR		
2	3	Cathy	Zulu	Finance		
3	4	David	Mokoena	Marketing		

Question 9

09 Find employees in IT, HR, or Finance using IN.

EXPECTED COLUMNS: id first_name last_name department

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace. default ▼

```
57
58 --Question 9
59 SELECT id,
60        first_name,
61        last_name,
62        department
63 FROM brightlearn.master_data.employees
64 WHERE department IN ('IT', 'HR', 'Finance');
65
```

Add parameter

Table ▼ +					Q 🔍	
	1 ² 3 id	A ^B C first_name	A ^B C last_name	A ^B C department		
1	1	Alice	Green	IT		
2	2	Brian	Lee	HR		
3	3	Cathy	Zulu	Finance		
4	5	Eva	Naidoo	IT		

Question 10

10 Find employees in IT with salary > 65000 and city Johannesburg.

EXPECTED COLUMNS: `id` `first_name` `last_name` `department` `salary` `city`

SQL code:

▶ Run all (1000) ✓ Just now (<1s) workspace. default ▾

```
66 --Question 10
67 SELECT id,
68        first_name,
69        last_name,
70        department,
71        salary
72 FROM brightlearn.master_data.employees
73 WHERE salary > 65000 AND city = 'Johannesburg';
74
```

Add parameter

Table ▾		+ 🔍 ⚙️ 📄 🗑️				
	id	first_name	last_name	department	salary	
1	1	Alice	Green	IT	70000	
2	5	Eva	Naidoo	IT	72000	